

Solution to the Daily Limit

Based on the problem provided in the image dailyintegral-limit-limit-1x1-2026-08-12.jpg, we are tasked with finding the limit of the following sequence as n approaches infinity:

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n \pmod{k}}{n^2}$$

1. Rewriting the Modulo Operator

To evaluate this limit, we can express the modulo operation in terms of the floor function. For any integers n and k with $k > 0$, the remainder of division can be written as:

$$n \pmod{k} = n - k \left\lfloor \frac{n}{k} \right\rfloor$$

Substituting this identity into our original summation yields:

$$S_n = \sum_{k=1}^n \frac{n - k \left\lfloor \frac{n}{k} \right\rfloor}{n^2} = \frac{1}{n^2} \sum_{k=1}^n n - \frac{1}{n^2} \sum_{k=1}^n k \left\lfloor \frac{n}{k} \right\rfloor$$

The first sum simplifies easily since we are summing the constant n exactly n times:

$$\frac{1}{n^2} \sum_{k=1}^n n = \frac{1}{n^2} (n \cdot n) = 1$$

So our limit expression becomes:

$$\lim_{n \rightarrow \infty} S_n = 1 - \lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n k \left\lfloor \frac{n}{k} \right\rfloor$$

2. Changing the Order of Summation

Let us focus on the remaining sum, which we will define as T_n :

$$T_n = \sum_{k=1}^n k \left\lfloor \frac{n}{k} \right\rfloor$$

Recognizing that $\left\lfloor \frac{n}{k} \right\rfloor$ represents the number of positive multiples of k that are less than or equal to n , we can rewrite this term as a sum of ones:

$$\left\lfloor \frac{n}{k} \right\rfloor = \sum_{m=1}^{\lfloor n/k \rfloor} 1$$

This allows us to express T_n as a double sum:

$$T_n = \sum_{k=1}^n k \sum_{m=1}^{\lfloor n/k \rfloor} 1 = \sum_{k=1}^n \sum_{m=1}^{\lfloor n/k \rfloor} k$$

The condition for the indices in the double sum is $1 \leq km \leq n$. By interchanging the order of summation—summing over m first—the bounds for k become $1 \leq k \leq \lfloor \frac{n}{m} \rfloor$:

$$T_n = \sum_{m=1}^n \sum_{k=1}^{\lfloor n/m \rfloor} k$$

Using the standard formula for the sum of the first N integers, $\sum_{k=1}^N k = \frac{N(N+1)}{2}$, we evaluate the inner sum:

$$T_n = \sum_{m=1}^n \frac{\lfloor \frac{n}{m} \rfloor (\lfloor \frac{n}{m} \rfloor + 1)}{2}$$

3. Evaluating the Asymptotic Behavior

We need to find the limit of $\frac{T_n}{n^2}$ as $n \rightarrow \infty$. We can write the floor function as $\lfloor \frac{n}{m} \rfloor = \frac{n}{m} - \{ \frac{n}{m} \}$, where $\{x\} \in [0, 1)$ is the fractional part of x . Thus, for large n , the dominant term in the numerator of our sum is:

$$\lfloor \frac{n}{m} \rfloor (\lfloor \frac{n}{m} \rfloor + 1) = \frac{n^2}{m^2} + O\left(\frac{n}{m}\right)$$

Dividing by $2n^2$, we approximate the terms of the sum:

$$\frac{1}{n^2} \frac{\lfloor \frac{n}{m} \rfloor (\lfloor \frac{n}{m} \rfloor + 1)}{2} = \frac{1}{2m^2} + O\left(\frac{1}{nm}\right)$$

Summing this from $m = 1$ to n gives:

$$\frac{T_n}{n^2} = \frac{1}{2} \sum_{m=1}^n \frac{1}{m^2} + O\left(\frac{1}{n} \sum_{m=1}^n \frac{1}{m}\right)$$

We know that the harmonic series grows logarithmically, meaning the error term $\frac{1}{n} \sum_{m=1}^n \frac{1}{m} = O\left(\frac{\ln n}{n}\right)$ vanishes as $n \rightarrow \infty$.

The primary term approaches half of the infinite sum of inverse squares, which is a well-known result from the Basel problem:

$$\lim_{n \rightarrow \infty} \sum_{m=1}^n \frac{1}{m^2} = \zeta(2) = \frac{\pi^2}{6}$$

Therefore, the limit of the second part of our original expression is:

$$\lim_{n \rightarrow \infty} \frac{T_n}{n^2} = \frac{1}{2} \left(\frac{\pi^2}{6} \right) = \frac{\pi^2}{12}$$

Conclusion

Putting it all together, we substitute this result back into our initial simplified limit equation:

$$\begin{aligned} \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n \pmod k}{n^2} &= 1 - \lim_{n \rightarrow \infty} \frac{T_n}{n^2} \\ &= 1 - \frac{\pi^2}{12} \end{aligned}$$

Thus, the final limit of the sequence is:

$$1 - \frac{\pi^2}{12}$$